ROCCO DI TELLA

rocco.di_tella@cfa.harvard.edu

EDUCATION

Universidad de Buenos Aires (UBA)
Licenciatura in Data Science (Joint B.S. + M.Sc.), GPA: 9.0/10 *2020 - 2024*
Thesis Advisors: Enzo Tagliazucchi, Gabriel Kreiman
Thesis project: Convolutional Neural Networks as models of human visual attention.

PhD in Computer Science *2026 - ongoing*
Advisors: Luciana Ferrer, Cecilia Garraffo

RESEARCH EXPERIENCE

Predoctoral Fellow — AstroAI, Harvard Center for Astrophysics *sept 2026 - present*
Advisor: Cecilia Garraffo. SAO Predoctoral Fellowship.
Generative AI architectures that respect known physical symmetries.

Postbaccalaureate Fellow — SPARK, Harvard Center for Astrophysics *sept 2025 - aug 2026*
Advisor: Juan Rafael Martínez Galarza.
Simulation-based inference to correct photon pile-up in X-ray observations of black-hole accretion.

Visiting Student — AstroAI, Harvard Center for Astrophysics *june - aug 2025*
Advisors: Cecilia Garraffo, Juan Rafael Martínez Galarza. SAO Visiting Student Fellowship.
Flow matching with transformers to generate multi-modal posteriors for galaxy images and X-ray spectra.

Research Assistant — Applied Artificial Intelligence Lab, UBA *jan - june 2025*
Advisor: Luciana Ferrer
Automatic detection of hallucinations in generative language models using grey-box calibration methods and white-box probing. Continued as a remote collaborator through Dec 2025, supported in part by the AISAR AI Safety Scholarship.

Research Assistant — Kreiman Lab, Harvard Medical School *2024 (remote)*
Advisor: Gabriel Kreiman
Licenciatura thesis co-advisor; collaboration continued after graduation.
Convolutional neural networks as models of human visual attention.

Research Assistant — Center for A.I. and Neuroscience, UTDT *2021 - 2022*
Developed Bayesian methods for efficient experiment design. Conducted data analysis and designed experiments in studies of human cooperation.

Summer Research Intern — A.I. and Robotics Lab (LINAR), Universidad de San Andrés *2021*
Deep learning for flight control automation.

ACADEMIC PUBLICATIONS

- **Di Tella R.**, Cuesta-Lazaro C., Böttger RL, Reeves J., Martínez-Galarza R.
"Probabilistic probes for galaxy evolution: signatures of AGN feedback in the AION foundation model", *NeurIPS 2026 Workshop on Interpretability for Discovery*
- **Di Tella R.***, Gago J.*, Shi J.
"Thinking out of distribution: what a PhD must train in the age of AI", *AJE Advances: Research in Epidemiology* (2026, in press)
- **Di Tella R.***, Böttger RL*, Cuesta-Lazaro C., Cappelluti N., Reeves J., Martínez-Galarza R.
"Exploring the connection between optical and X-ray", *Conference on Physics and AI (PAI26)*, Stanford University (2026)
- Pinciroli N., **Di Tella R.**, Cuesta-Lazaro C, Smith MJ, Garraffo C, Martínez-Galarza R.
"Augmenting Representations with Scientific Papers", *Neural Computing and Applications* (2026)

- Borrell T., Dillmann S., Duraphe K., Eris F., Iyer K., Khederlarian A., Kumar A., Marraffini G., Smith MJ, Sourav S., **Di Tella R.**, Wu JF (alphabetical order)
"The Platonic Universe: Do Foundation Models See the Same Sky?", *ICML 2026 Workshop on Mechanistic Interpretability*; arXiv:2509.19453

- Pérez-Carrasco MI, Park CF, Zhu Q, **Di Tella R.**, Ayazpour Z, Abad GG, Garraffo C
"Reconstruction Is Not Enough: Evaluating Self-Supervised Objectives for Spectroscopic Satellite Embeddings", *ICLR 2026 Workshop on Machine Learning for Remote Sensing*

- Rieznik A., **Di Tella R.**, Schvartzman L., Babino A. (2021)
"Optimum Integration Procedure for Connectionist and Dynamic Field Equations", *Frontiers in Neurorobotics*

- Cavanna F., Pallavicini C., Milano V., Cuiule J., **Di Tella R.**, González P., Tagliazucchi E. (2021)
"Lifetime Use of Psychedelics is Associated With Better Mental Health Indicators During the COVID-19 Pandemic", *Journal of Psychedelic Studies*

*Equal contribution.

TALKS & POSTERS

- "Exploring the connection between optical and X-ray emissions in Active Galactic Nuclei", AstroAI Seminar, Center for Astrophysics, 2026 (talk).

- Tutorial on agentic AI for researchers, AstroAI Workshop, CfA, 2026 (invited tutorial speaker).

- "The Platonic Universe: Do Foundation Models See the Same Sky?", AstroAI Workshop, CfA, 2026 (poster and short talk).

- "The Platonic Universe: Do Foundation Models See the Same Sky?", New England Mechanistic Interpretability Workshop (NEMI), 2026 (poster).

- "Augmenting Representations with Scientific Papers", ICLR 2026 Workshop on Foundation Models for Science (poster).

- "Where's Waldo? How Human Brains and Neural Networks Look for Stuff Differently", AstroAI Seminar, Center for Astrophysics, 2025 (talk).

- "Modeling X-ray photon pile-up with a normalizing flow", NeurIPS, 2025 (poster).

- "Building Honest Agents through introspection", AstroAI workshop, 2025 (poster).

- "Finding Waldo", Simposio Científico de Inteligencia Artificial y Aplicaciones, 2024 (poster).

- "A Bayesian Look", Bayes Plurinacional Workshop, 2024 (short talk and poster).

- "Fast Nonparametric Bayesian Framework for On-the-Fly Adaptive Design Optimization", Bayes Plurinacional Workshop, 2023 (poster).

- "A Massive Experiment on Choice Blindness", Sociedad Argentina de Neurociencia, 2017 (poster).

TEACHING EXPERIENCE

Teaching Assistant — Universidad de Buenos Aires (UBA) *aug 2025*

- Intensive winter course on applications of AI for astrophysics

Teaching Fellow — Universidad de San Andrés (UdeSA) *feb - july 2025*

- Advanced undergraduate course on Machine Learning

Teaching Assistant — Universidad de Buenos Aires (UBA) *jan - dec 2024*

- Helped launch two new graduate courses: Bayesian Inference (taught half of the theory lectures) and Causal Inference
- Advanced undergraduate course on Statistics

Teaching Assistant — Universidad Torcuato Di Tella (UTDT) *2022 and 2024*

- Undergraduate course on Cognitive Science
- Undergraduate course on Statistics

FELLOWSHIPS & SCHOLARSHIPS

SPARK Postbaccalaureate Fellowship — Center for Astrophysics, Harvard & Smithsonian *2025*
Twelve months of funded full-time research.

Visiting Student Fellowship — Smithsonian Astrophysical Observatory *2025*
Ten-week research visit at AstroAI.

AI Safety Scholarship — AISAR (AI Safety Argentina) *2025*
Six-month research stipend and compute.

PROGRAMS & SERVICE

CAMBRIA — Cambridge Boston Alignment Initiative *2026*
Three-week program on mechanistic interpretability and alignment.

Reviewer — Workshop on Representations for the Physical Sciences *2026*

INDUSTRY & OUTREACH

Editorial Board Member — Ciencia Hoy *nov 2025 - present*
Popular-science magazine of the Universidad Nacional de Tres de Febrero (UNTREF).

Data Scientist — GRIDX Biotech Venture Capital *2021 - 2022*
Started a data consulting team which grew into an independent company. Financial forecasting and consulting for portfolio companies.

Data Scientist — El Gato y La Caja *2017 - 2021*
Engaged in data storytelling aimed at a general audience and designed experiments for cognitive science research. Click here to see showcased work.

Science Communicator — Instituto Baikal *2019 - 2020*
Coordinated "Frontiers of Science", a seminar series for a general audience.

SKILLS

Methods: simulation-based inference, normalizing flows and flow matching, probing and sparse autoencoders, Bayesian and causal inference, agentic AI workflows for research
Tools: Python (PyTorch, JAX), R, Julia